

When Should Financial QA Be Agentic?

Selective Orchestration for Accuracy–Cost-Efficient Financial Reasoning

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Abstract

Iterative language-model agents can improve financial question answering (QA), but they also spend inference on questions that compact evidence-conditioned readers may already solve. We study *when* financial QA should be agentic rather than treating iterative reasoning as a uniform default. We introduce Selective Finance Nexus, a deterministic-first workflow that constructs a compact evidence dossier and uses a frozen, pre-generation risk gate to choose between one-call adjudication and a bounded ReAct process. The evaluation deliberately controls evidence access: it isolates reasoning and orchestration from end-to-end retrieval across FinanceBench, FinDER, FinQA, TAT-QA, and ConvFinQA. Existing pilot examples are used only for development; the final protocol contains 900 disjoint examples, leakage tests, dataset-native scoring for numerical tasks, and provider-level measurements of tokens, dollars, calls, and latency. [TBD from frozen artifact: primary final manifest and paired statistics] We report narrative benchmarks separately because their model-judge scores are not human-validated. [TBD from frozen artifact: frozen router operating point, accuracy–cost trade-off, and cross-family replication] These results test a practical alternative to uniformly agentic finance systems: reserve iterative computation for questions whose observable evidence and question features predict that it is useful.

Keywords

financial question answering, language-model agents, selective computation, cost-aware reasoning, ReAct

1 Introduction

Financial QA combines requirements that are individually difficult and jointly unforgiving: locating the relevant period and entity, interpreting tables and prose, executing arithmetic, preserving units and signs, and grounding narrative conclusions in filings. Iterative language-model agents are a natural response. A ReAct-style system can search, inspect evidence, revise a hypothesis, and stop when it has an answer. That flexibility, however, has a direct operational cost. Every additional step consumes tokens and latency, and additional reasoning can also introduce a new arithmetic, scale, or evidence-selection error into an otherwise straightforward question.

This paper asks a narrower question than whether agents can answer financial questions: *when is iterative agency worth its cost?* We study this question under controlled evidence access. Each system receives the same benchmark-provided, leakage-audited evidence corpus and the same permitted tools. This design does not measure end-to-end retrieval. Instead, it makes the orchestration decision identifiable: differences arise from how a system selects,

organizes, and reasons over available evidence rather than from access to different documents.

Our method, *Selective Finance Nexus*, begins with deterministic query analysis and evidence selection. A frozen gate then routes each example, before any answer generation, to either a one-call structured adjudicator or a bounded iterative ReAct process. The gate uses only observable, target-independent features of the question and evidence, such as length, numerical and temporal cues, lexical coverage, retrieval-score margin, and evidence dispersion. It is trained only on a development partition to predict the asymmetric event that ReAct succeeds while the static workflow fails. This formulation targets incremental value rather than generic difficulty.

We organize the study around four research questions:

RQ1. Does deterministic-first orchestration improve the quality–cost frontier over direct prompting, structured one-call reasoning, and uniformly applied ReAct?

RQ2. Which observable financial-question characteristics distinguish cases helped by iterative agency from cases solved more efficiently by a static workflow?

RQ3. Can a development-frozen gate preserve the stronger system’s accuracy while avoiding unnecessary iterative calls?

RQ4. Are the conclusions stable across datasets, model families, and dataset-native evaluation protocols?

The study makes three contributions. First, it defines a selective orchestration method in which deterministic components are workflow stages, not separately personified agents, and escalation occurs before answer generation. Second, it establishes a leakage-audited, paired protocol spanning numerical, tabular, conversational, and narrative financial QA. The earlier 250-example sweep is used only for method development; all reported confirmatory results come from a disjoint frozen manifest. Third, it evaluates efficiency at provider level—tokens, dollar cost, calls, and latency—and reports an offline oracle router to quantify the remaining headroom for selection. We treat narrative model-judge results as secondary sensitivity analyses and do not make an end-to-end RAG claim.

[TBD from frozen artifact: headline finding from the v2 final analysis]

2 Related Work

Financial QA. FinQA introduced expert-authored financial-report questions with executable programs; TAT-QA broadened the evidence to hybrid tables and text, and ConvFinQA added conversational dependencies [3, 4, 18]. FinanceBench instead evaluates open-book questions over filings, and FinDER emphasizes terse professional queries paired with expert evidence for retrieval-augmented generation [5, 11]. FinQAPT joins report identification, context extraction, and answering in an end-to-end pipeline [13]. Our

controlled-evidence setting holds retrieval fixed to isolate reasoning orchestration. Accordingly, FinDER and FinanceBench serve only as secondary semantic stress tests rather than being pooled with execution-scored numerical tasks.

Agentic financial reasoning and evaluation. ReAct interleaves language-model reasoning with external actions [17]. In finance, multi-agent reflection improves answers through one or more critic roles [9], whereas FinAgentBench evaluates two-stage agentic retrieval and FinResearchBench evaluates long-form research agents with logic-tree judgments [6, 14]. These works motivate iterative computation, but do not establish that every question benefits from it. Indeed, both general reasoning and financial sentiment studies document overthinking or diminishing returns from extra reasoning [2, 15].

Selective and cost-aware inference. Selective prediction formalizes risk–coverage trade-offs by declining or escalating uncertain cases [10]. FrugalGPT and RouteLLM learn cost-sensitive cascades or routers across models [1, 12]; recent agent-efficiency work likewise argues for joint effectiveness and operational-cost measurement [7, 16]. We apply this principle within a fixed model and evidence budget: a pre-generation router chooses between deterministic-first, one-call adjudication and iterative ReAct. We therefore compare quality jointly with calls, tokens, dollar cost, and latency, and use RAGAS-style semantic judgments only as a secondary, judge-sensitive analysis [8].

3 Selective Financial QA

Let q be a question and $\mathcal{E}$ the controlled evidence corpus supplied for that example. All compared systems operate on the same $(q, \mathcal{E})$ and share the same evidence and answer budgets described in Section 4. We compare two complementary policies: a static deterministic-first workflow S and an iterative workflow R . A selector g_τ chooses one policy before either generates an answer:

$$A(q, \mathcal{E}) = \begin{cases} R(q, \mathcal{E}) & \text{if } g(z(q, \mathcal{E})) \geq \tau, \\ S(q, \mathcal{E}) & \text{otherwise,} \end{cases} \quad (1)$$

where z contains no answer, gold program, answer type, scale, derivation, or other target-derived field.

3.1 Static deterministic-first workflow

The static path has four stages. First, a deterministic parser identifies lexical indicators for entities, reporting periods, quantities, operators, units, table structure, conversational context, and narrative implication. Second, it creates several fixed query views of q , searches only within $\mathcal{E}$, and deduplicates the returned chunks. Third, it ranks chunks by lexical relevance and assembles a dossier under a fixed token budget. Fourth, a single language-model call produces a concise answer with an evidence-grounded calculation or rationale. A deterministic validator normalizes output formatting and flags malformed units, signs, percentages, or unsupported answers; the validator never consults the reference answer. Its contribution is isolated in an ablation.

3.2 Iterative workflow

The iterative path follows a bounded ReAct interface over the same corpus. At each step the model may search the controlled evidence,

Method diagram placeholder

Controlled evidence $\rightarrow$ deterministic features and selection $\rightarrow$ frozen pre-router $\rightarrow$ {one-call adjudication, bounded ReAct}. Replace from the frozen architecture-figure source; show that routing precedes generation.

Figure 1: Selective Finance Nexus. Deterministic components are workflow stages rather than autonomous agents. Both paths receive the same controlled evidence and tools.

inspect a matched passage, or finish with an answer. The run terminates after at most seven model steps. Search and lookup operations, evidence returned, truncation, and all model usage are recorded. ReAct therefore has flexibility to reformulate queries or repair intermediate reasoning, but it cannot access evidence or tools unavailable to the static path.

3.3 Pre-generation selector

The selector uses deterministic features available before generation: question and evidence lengths; counts of numbers, operators, periods, and entities; table and conversational indicators; narrative and implication cues; lexical evidence coverage; the top retrieval score and top-one/top-two margin; chunk count; and evidence-score dispersion. We standardize numeric features and fit an L2-regularized logistic regression with balanced class weights and random seed 20260709. The development target is one exactly when R is correct and S is incorrect. Dataset identity is excluded so the selector cannot memorize dataset-level policy.

Five-fold stratified cross-validation chooses the least expensive threshold whose accuracy is within one absolute percentage point of the better development accuracy of S and R . If a fold contains fewer than five positive routing examples or cross-validated AUROC is below 0.60, the preregistered fallback gate escalates questions with narrative/implication cues, lexical coverage below the development median, or retrieval margin in the lowest development quartile. The selected feature transform, model coefficients or fallback thresholds, and operating threshold are hashed and frozen before evaluation.

3.4 Comparators and oracle headroom

Direct prompting performs one question search and one answer call. The structured one-call baseline uses the same dossier and input budget as the static path but prompts explicitly for a calculation or short reasoning trace. Static Finance Nexus is S without routing; ReAct is R on every example; Selective Finance Nexus applies g_τ ; and an offline oracle chooses the correct outcome whenever either S or R succeeds. The oracle is an analysis upper bound, not a deployable system.

4 Experimental Protocol

4.1 Scope, datasets, and split discipline

The evaluation is evidence-conditioned. For FinanceBench and FinDER we expose only benchmark evidence passages; for FinQA,

TAT-QA, and ConvFinQA we expose their legitimate tables, text, and dialogue context. Gold answers and target-derived annotations are never model-visible. In particular, FinanceBench justifications and TAT-QA derivations and gold scales are removed. The resulting experiment isolates orchestration over controlled evidence and must not be interpreted as end-to-end retrieval or RAG performance.

The 50-example-per-dataset sweep used during method construction (250 examples total) is the complete development set. The final manifest contains 900 disjoint examples: the 100 unused FinanceBench examples; 200 stratified FinDER examples; 200 fresh FinQA validation examples; 200 TAT-QA validation examples stratified by answer type and evidence source; and 200 ConvFinQA examples whose dialogue identifiers do not occur in development. Manifest tests enforce empty development/final intersections and dialogue-level separation. Dataset revisions, row identifiers, content fingerprints, prompts, scorers, router artifacts, model settings, and price snapshots are recorded under protocol `finance_icaif26_v2`. No prompt, routing, or scoring rule is changed after the final manifest is opened.

4.2 Systems, models, and budgets

We evaluate Direct, structured one-call reasoning (CoT/PoT), ReAct, Static Finance Nexus, and Selective Finance Nexus. Direct, CoT/PoT, and Static Finance Nexus receive the same maximum dossier-token budget; all systems search the same evidence corpus and have identical calculator/tool permissions. ReAct is capped at seven model steps. Runs are paired by example and resumed only after a typed infrastructure failure, never in response to an incorrect valid output.

The primary full evaluation uses google/gemini-2.5-flash with temperature zero and model thinking disabled. A fixed 50-example-per-dataset subset is replicated with openai/gpt-5.6-luna and reasoning effort disabled. The frozen protocol records the provider-resolved model identifier, evaluation date, and request parameters rather than relying on a mutable alias. Each response records input, cached, reasoning, output, and total tokens; model and retrieval calls; provider-reported or price-manifest-estimated dollar cost; retries; and end-to-end latency. The full study has a preregistered API-spend ceiling of US\$150.

4.3 Metrics and inference

The primary objective endpoints are FinQA execution accuracy, the official TAT-QA exact/F1 measures with prediction-side scale inference, and ConvFinQA execution accuracy. The headline quality measure is the unweighted macro-average across these three datasets. We report every dataset separately, paired bootstrap 95% confidence intervals, exact McNemar tests for binary correctness, and Holm-adjusted dataset-level p -values.

FinanceBench strict exact/numeric accuracy and FinDER semantic correctness/faithfulness are secondary stress tests. For non-exact narrative cases, two independently prompted judges—the frozen GPT-5.6 Terra and Gemini 2.5 Pro configurations—receive blinded, randomized system outputs. We report judges separately together with correlation and disagreement; without human validation, these scores do not enter the headline macro-average and cannot support a universal narrative-quality claim.

Efficiency endpoints are model and retrieval calls, token categories, dollars per example, median and 95th-percentile latency, accuracy at equal dollar cost, and cost at equal accuracy. Paired bootstrap intervals quantify cost differences. Threshold sweeps yield the selective accuracy–cost frontier, while static-path, no-validator, and no-router ablations isolate each architectural contribution.

4.4 Validity and reproducibility controls

Automated tests scan rendered prompts and evidence for forbidden target fields, confirm deterministic manifest regeneration, exercise scale/sign/percentage scoring fixtures, verify that command-line model identifiers reach provider requests, and ensure resumption cannot duplicate a completed framework/example pair. Every result row links to hashes of the protocol, prompt, evidence, scorer, and router artifacts. Infrastructure failures remain distinct from model nontermination or malformed answers. *[TBD from frozen artifact: protocol freeze hash, dataset fingerprints, and telemetry-completeness rate]*

5 Results

This section is intentionally unpopulated until all five systems complete the immutable final manifest. Development-sweep values are not carried into confirmatory tables. Table 1 will report native quality metrics and measured efficiency from the single frozen analysis artifact.

RQ1: quality–cost frontier. The paired estimate is not yet populated. *[TBD from frozen artifact: primary comparison, intervals, and equal-cost analysis]* The claim is accepted only if Selective Nexus either improves quality while reducing cost or falls within one absolute point of the best primary macro score while reducing dollar cost by at least 30%.

Secondary stress tests. *[TBD from frozen artifact: FinanceBench strict score and two-judge FinDER/FinanceBench sensitivity results]* We will state disagreements between semantic judges rather than averaging them into a more decisive-looking score.

RQ4: cross-family stability. *[TBD from frozen artifact: fixed 250-example GPT-5.6 Luna replication with paired uncertainty]* The paper will distinguish directional replication from a fully powered second-model evaluation.

Ablations. *[TBD from frozen artifact: no structured selection, no validator, and no-router ablations]* The final text will attribute gains only to components with paired evidence and will report any dataset-specific regression exceeding five points.

6 When Does Agency Help?

6.1 Routing behavior

Figure 2 will plot all development-frozen routing thresholds and mark the preregistered operating point; the final-test point is evaluated once and is not used to move the threshold. We compare this point with always-static, always-ReAct, the rule-gate fallback when applicable, and the offline oracle.

[TBD from frozen artifact: selector AUROC or fallback activation, chosen threshold, escalation rate, calibration, and oracle headroom] To answer RQ2 without post-hoc storytelling, feature effects are

Table 1: Primary held-out results. Quality is the unweighted macro-average of FinQA execution accuracy, TAT-QA official score, and ConvFinQA execution accuracy. Cost and latency are measured, not inferred from call counts. Best results and statistically indistinguishable results will be marked only after paired testing.

System	FinQA	TAT	Conv	Macro	Calls	Tok.	Cost	P95
Direct	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD
CoT/PoT	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD
ReAct	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD
Static Nexus	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD
Selective Nexus	TBD	TBD	TBD	TBD	TBD	TBD	TBD	TBD

TAT: TAT-QA official score; Conv: ConvFinQA execution accuracy; Macro: unweighted primary macro; Calls and Tok.: per example; Cost: US\$/example; P95: end-to-end latency. Each TBD cell names its frozen source in the L²J_X source and must be replaced before submission.

Accuracy–cost Pareto plot placeholder

Replace from the frozen threshold-sweep artifact. Plot primary macro quality against measured US\$/example; label Static, Selective, ReAct, and Oracle.

Figure 2: Quality–cost frontier over frozen routing thresholds. Error bars are paired bootstrap 95% confidence intervals.

Table 2: Routing and failure-analysis template. R+ denotes ReAct-only correctness, S+ static-only correctness, and BW both wrong. Counts come only from the frozen disagreement artifact.

Trace-observable category	R+	S+	BW
Evidence selection/truncation	TBD	TBD	TBD
Arithmetic/program	TBD	TBD	TBD
Unit/scale/sign	TBD	TBD	TBD
Entity/period/dialogue	TBD	TBD	TBD
Narrative support	TBD	TBD	TBD
Format/nontermination	TBD	TBD	TBD

reported with cross-validated uncertainty and are interpreted as routing associations, not causal properties of financial questions.

6.2 Disagreement and failure analysis

We derive failure labels from trace-observable events and scorer outputs rather than unblinded framework narratives. The analysis distinguishes evidence selection/truncation, arithmetic or program execution, scale/unit/sign, entity or period mismatch, conversational-state loss, unsupported narrative inference, malformed or nonterminating output, and scorer disagreement. Because no human annotation is available, narrative categories are explicitly exploratory and semantic-judge conflicts remain unresolved rather than adjudicated by the authors.

[TBD from frozen artifact: one-sided disagreement rates and trace-based error distribution] We will not convert judge-dependent narrative labels into a quantitative causal claim.

7 Limitations and Threats to Validity

Controlled rather than end-to-end evidence. The study measures orchestration conditional on benchmark-provided evidence. It cannot establish how the systems retrieve from complete filings, handle OCR, or operate under missing evidence. Calling the primary result “RAG performance” would therefore be incorrect.

Narrative evaluation. FinanceBench and especially FinDER contain semantically valid answers that strict matching may miss. Two model judges provide sensitivity analyses, but no human labels validate their calibration. Judge agreement does not remove shared model bias. These datasets remain secondary and do not enter the headline macro-average.

Development size and selection shift. The selector is learned from 250 examples spanning five benchmarks. Rare modes of ReAct-only success may be poorly estimated, and balanced logistic regression does not guarantee calibrated probabilities under a new question distribution. The rule-gate fallback and oracle establish bounds, not a remedy for distribution shift.

Model and price dependence. Provider behavior, latency, and prices change. We snapshot resolved identifiers and prices, report raw token counts alongside dollars, and replicate on a second model family, but conclusions may not transfer to other models or deployment settings. Latency also reflects evaluation-time provider load.

Benchmark and scoring validity. Dataset-native execution metrics emphasize answer correctness but may under-reward valid explanations. Conversely, tolerance, scale inference, and normalization can create false matches. Frozen fixtures, paired tests, and separate semantic sensitivity analyses reduce but do not eliminate this risk. The five benchmarks are public English-language tasks and do not cover private data, multilingual reporting, or investment-decision safety.

No autonomous multi-agent claim. The parser, evidence selector, dossier builder, and validator are deterministic workflow components. Describing each as an autonomous agent would inflate the architectural claim. Our contribution is selective orchestration between a static language-model call and bounded iterative reasoning.

8 Conclusion

This work reframes agentic financial QA as a resource-allocation problem. Selective Finance Nexus uses target-independent, pre-generation signals to reserve bounded ReAct reasoning for cases predicted to benefit, while handling other questions with deterministic evidence organization and one language-model adjudication call. A leakage-audited, disjoint protocol measures this choice under controlled evidence using native quality metrics and provider-level cost. [TBD from frozen artifact: final conclusion supported by the primary paired analysis] Regardless of the direction of the final comparison, the controlled design separates a question often blurred in agent evaluations: whether better document retrieval is needed, or whether the available evidence simply needs a different amount of inference.

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